

YORKTOWN USCG, VA, USA

WMO: 997315

Lat: 37.2265N

Lon: 76.4788W

Elev: 10

StdP: 101.20

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -6.1	(c) -3.8	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 14.4	(k) 3.0	(l) 12.5	(m) 2.6	(n) 5.2	(o) 340	(p) 0.500

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 6.4	(c) 32.8	(d) N/A	(e) 31.6	(f) N/A	(g) 30.4	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 3.9	(p) 240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 12.4	(b) 10.7	(c) 9.0	(e) -7.7	(f) 35.7	(g) 3.0	(h) 1.7	(i) -9.9	(j) 36.9	(k) -11.6	(l) 37.8	(m) -13.3	(n) 38.8	(o) -15.5	(p) 40.0		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.0	4.8	5.7	9.1	15.4	20.0	24.5	26.5	25.8	22.7	17.4	11.2	7.9
	DBStd	8.69	5.54	5.06	4.96	4.51	3.79	2.84	2.12	1.96	2.70	3.97	4.16	4.81
	HDD10.0	544	181	140	81	7	0	0	0	0	0	2	35	99
	HDD18.3	1811	421	355	289	112	27	1	0	0	2	63	216	325
	CDD10.0	2722	18	19	52	168	311	434	512	491	381	232	70	34
	CDD18.3	947	0	1	2	23	80	185	253	232	133	36	1	1
	CDH23.3	7096	0	1	14	149	433	1412	2371	1917	674	123	1	1
	CDH26.7	1972	0	0	1	28	98	422	776	509	124	14	0	0
Wind	WSAvg	4.4	4.6	4.8	5.2	5.0	4.5	4.0	3.8	3.8	4.2	4.5	4.5	4.3
Precipitation	PrecAvg	1167	83	76	95	79	89	98	138	128	131	95	73	86
	PrecMax	1589	145	238	242	142	130	229	226	295	446	216	266	177
	PrecMin	773	33	30	17	12	30	42	24	28	28	6	2	32
	PrecStd	208	32	43	43	32	25	45	54	72	95	52	57	37
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.0	22.8	25.4	29.6	31.3	34.4	35.4	33.9	31.7	28.4	23.3	22.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	18.8	19.6	22.3	27.0	29.4	32.0	33.0	31.9	29.8	26.0	21.1	20.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	16.4	17.5	19.6	25.0	27.5	30.5	31.7	30.7	28.0	24.6	19.1	18.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	13.8	14.0	17.3	22.6	25.7	29.1	30.5	29.4	26.7	23.2	17.5	15.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	7.1	7.4	7.5	7.9	6.8	6.4	6.4	5.7	5.3	6.1	6.7	6.8
		MCDBR	11.1	11.3	12.0	10.7	9.1	8.6	8.4	7.6	7.3	7.6	9.1	10.0
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.306	0.317	0.345	0.392	0.434	0.485	0.515	0.511	0.410	0.353	0.322	0.309	
	taud	2.542	2.504	2.459	2.336	2.249	2.157	2.088	2.088	2.357	2.516	2.554	2.564	
	Ebn at Noon	890	920	921	889	853	807	778	772	841	868	863	861	
	Edh at Noon	77	90	104	124	137	151	160	157	114	88	76	71	
All-Sky Solar Radiation	RadAvg	2.33	3.08	4.11	5.21	5.89	6.35	6.05	5.46	4.43	3.50	2.63	2.07	
	RadStd	0.19	0.34	0.33	0.38	0.53	0.43	0.29	0.28	0.38	0.41	0.22	0.16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (71 neighbors)		+0.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+161	+104

Nomenclature: See separate page